

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #57 — Pet Adoption & Shelter Management System

Actors: Adoption Applicant, Shelter Staff, Veterinarian, Payment Gateway (external system)

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	
FR-001	Functional	The system shall allow Shelter Staff to log an animal's vaccination and medical intake records, updating the animal's availability status to 'Ready for Adoption'.	High	Pass: an adoption application can only be linked to an animal record marked 'Ready for Adoption'. Fail: an animal under medical quarantine is marked as adopted.	Prevents unsafe or premature adoption of an animal that has not cleared a medical screening.	
FR-002	Functional	The system shall allow an Adoption Applicant to submit a screening questionnaire (housing type, prior pet experience, household members) for a specific animal, which Shelter Staff can review and approve or reject.	High	Pass: every submitted questionnaire is stored with a status of Pending, Approved, or Rejected and is linked to both the applicant and the target animal. Fail: an application is accepted with no linked questionnaire record.	Ensures adopter suitability is assessed before an animal is placed with them.	
FR-003	Functional	The system shall allow an Adoption Applicant to search and filter the pet catalogue by attributes including Species, Age, Breed, and Good with Kids.	High	Pass: every result returned matches all attributes selected in the filter. Fail: the result set includes an animal that does not match one or more selected filters.	Helps applicants find a suitable animal quickly, which improves adoption throughput.	
FR-004	Functional	The system shall allow Shelter Staff to assign an animal to a foster home and record the placement's start and end dates.	Medium	Pass: the animal's record shows its current foster assignment and duration whenever one is active. Fail: an actively fostered animal is shown as 'in shelter' with no foster record.	Keeps the animal's physical location and welfare responsibility accurately tracked.	
FR-005	Functional	The system shall allow Shelter Staff to finalize an approved adoption, which processes the adoption fee payment, updates the animal's status to 'Adopted', and removes the animal from the active search catalogue.	High	Pass: once finalized, the animal no longer appears in active search results and a payment confirmation is recorded against the adoption. Fail: an adopted animal is still listed as available, or the adoption is finalized with no payment record.	Keeps the catalogue accurate and prevents duplicate applications on an animal that is already adopted.	
NFR-001	Nonfunctional	The pet search catalogue must support multi-attribute filtering (Species, Age, Breed, Good with Kids) with a page response time under 200 ms.	High	Pass: benchmarking tests confirm the target latency and security standards are met under simulated peak load.	Ensures the search experience stays responsive during high-traffic periods (e.g., after a shelter promotion).	
NFR-002	Nonfunctional	Adoption fee payments processed through the external Payment Gateway shall be transmitted over an encrypted (TLS 1.2 or higher) connection, and no raw payment card data shall be stored on shelter servers.	High	Pass: a security scan confirms all payment traffic uses TLS 1.2+ and the shelter database contains no stored card numbers. Fail: any payment request is sent unencrypted or card data is found at rest.	Protects applicant financial data and keeps the shelter out of PCI-DSS scope for card storage.	